A-A-51143A
July 24, 1990
SUPERSEDING
A-A-51143
March 29, 1985

COMMERCIAL ITEM DESCRIPTION

CUTTER BIT (CARBIDE TIP)

The General Services Administration has authorized the use of this commercial item description.

Abstract. This description covers cutter bits with brazed sintered carbide tips that are supported by a tool holder and used on metal cutting machines for taking roughing to finishing cuts on cast iron, steel, non-ferrous, and non-metallic materials.

Classification. The cutter bits shall be of the following styles, classes, and sizes. The style, class, and size shall be as specified.

Style - AR - 0 degree side cutting edge angle, right hand.
Style - AL - 0 degree side cutting edge angle, left hand.
Style - BR - 15 degree side cutting edge angle, right hand.
Style - BL - 15 degree side cutting edge angle, left hand.

Class - The class of carbide tool specified shall be as selected from table I according to typical use.

Size 04 - 1/4 inch wide x 1/4 inch high.
Size 05 - 5/16 inch wide x 5/16 inch high.
Size 06 - 3/8 inch wide x 3/8 inch high.
Size 07 - 7/16 inch wide x 7/16 inch high.
Size 08 - 1/2 inch wide x 1/2 inch high.
Size 10 - 5/8 inch wide x 5/8 inch high.
Size 12 - 3/4 inch wide x 3/4 inch high.

Beneficial comments recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis TN 38114-5051.

AMSC N/A FSC 3455

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TABLE I. Intended uses, typical machining applications.

Class	Commercial designation grade	*Typical machining operations	*Typical materials	Typical cuts	
				Cutting speed	Tool feed
1	C-1	Heavy duty roughing cuts by turning, milling, shaping and planing under favorable machining conditions	Low hardness gray cast iron, metal that generate short chips, non-ferous metals, low tensile strength steel, stainless steel, non-metallic materials	Low	Coarse
2	C-2	Semi-roughing and general purpose machining by turning, milling, shaping, planing, and boring	Gray cast irons having hardnesses up to Rockwell C 20 non-ferrous metals, titanium metals, high temperature steel alloys, monel, bronze, alloys, brasses	Medium	Medium
3	C-3	Semi-finishing and finishing cuts by turning, milling, drilling, boring, and broaching	Gray cast irons over Rockwell C 20 hardness, malleable iron and modular iron that generates short chips, hardened steel, silicon aluminum alloys, titanium metals, plastics, hard rubber, fiberglass	Medium to high	Medium to fine
4	C-4	Finishing cuts, precise dimension control. and best surface finishes by turning, boring, and milling	Very hard gray iron castings and chilled castings harder than Rockwell C 60, high silicon aluminum alloys, hardened steel, highly abrasive plastics, phenolic laminates, ceramics	High	Fine
5	C-5	Heavy duty roughing cuts by turning, milling, shaping, and planing under unfavorable machining conditions	Carbon and alloy steel, steel castings with cored sections, cavities, sand inclusions, malleable iron that generates relatively long chips	Medium to low	Medium to coarse
6	C-6	Semi-roughing and general purpose machining by turning, milling, shaping and planing	Carbon and alloy steels, steel castings, malleable iron that generates relatively long chips	Medium	Medium
7	C-7	Semi-finishing and finishing cuts by turning, milling, threading and boring	Carbon and alloy steels, steel castings, stainless steels, malleable and modular irons that generate relatively large chips	Medium to high	Medium to fine
8	C-8	Fine finishing cuts, fine surface finishes, and precise dimension control by turning and boring. Precise threading by high speed, single point threading machines	Carbon and alloy steels, steel castings	High	Fine

*Note: "Typical operations" and "Typical materials" are listed in the approximate order of decreasing frequency of usage, i.e. most frequent applications are listed first followed by less frequent applications.

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Salient characteristics.

1. Design. The cutter bit style and size shall conform to the requirements ANSI B212.1 "Carbide Blanks Brazed and Solid Single-Point Tools."

2. Dimensions. The dimensions of the cutter bits shall conform to table II for the size required for a specific procurement. The tolerances for all dimensional data shall be the manufacturers standard as specified on their part drawings.

3. Material. The tool body shall be steel with a brazed sintered carbide tip.

4. Marking. The body of the tool shall be permanently marked with the manufacturer's name or trade mark.

TABLE II. Cutter bit sizes and dimensions.

Tool dimensions, inches					Tip dimensions, inches			
Size	Width	Height	Length	Nose radius	Tip overhang	Thick	Width	Length
04	1/4	1/4	2	1/64	1/32	3/32	3/16	5/16
05	5/16	5/16	2-1/4	1/64	1/32	3/32	1/4	1/2
06	3/8	3/8	2-1/2	1/64	1/32	3/32	1/4	1/2
07	7/16	7/16	3	1/32	1/32	3/32	1/4	1/2
08	1/2	1/2	3-1/2	1/32	1/32	1/8	5/16	5/8
10	5/8	5/8	4	1/32	1/32	5/32	3/8	3/4
12	3/4	3/4	4-1/2	1/32	1/32	3/16	7/16	13/16

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition, without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Environmental protection. During the manufacture of the cutter bit, materials hazardous to the ecological system as prohibited by the Federal, state, or

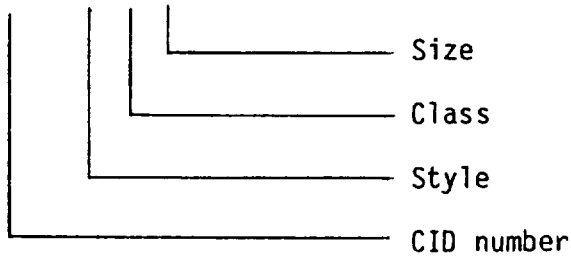
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local statutes in effect on the date of the contract shall not be used or emitted.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract order.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable Arabic numbers shall be substituted for Roman numerals.

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Quality assurance.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

Responsibility for compliance. All items shall meet the requirements specified herein. The inspection set forth in this description shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the description shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Sampling inspection. As a minimum sampling inspection, the supplier shall randomly select the number of samples in table III that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination, the test, and the inspection. Should any of the sample items fail to meet either the examination or

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inspection below, the entire lot shall be considered defective and unsuitable for acceptance. A lot is defined as all items presented at one time for acceptance.

TABLE III. Sampling plan.

Lot size	Sample size	Lot size	Sample size
2 to 8	2	151 to 280	32
9 to 15	3	281 to 500	50
16 to 25	5	501 to 1,200	80
26 to 50	8	1201 to 3,200	125
51 to 90	13	3201 to 10,000	200
91 to 150	20		

Examination. Each cutter bit submitted for sample inspection shall be visually examined to determine conformance with all requirements of this specification.

NOTES.

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Style, class, size, and quantity required.

Federal Standard FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

ANSI B212.1 is available from American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.

ASTM D 3951 is available from American Society for Testing and Materials, 1916 Race street, Philadelphia, PA 19103-1187.

Cross reference of classification changes. Classification changes from the previous issues(s) are as follows:

A-A-51143	A-A-51143A
Size 16	Deleted
Size 16A	Deleted

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TABLE III
TABLE IV

Deleted
TABLE III

MILITARY INTEREST:

Custodians

Army - AL
Air Force - 99

Review Activities

Army - 84
DLA - GS

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3455-0133

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-51143A	2. DOCUMENT DATE (YYMMDD) 90 07 24

3. DOCUMENT TITLE
CUTTER BIT (CARBIDE TIP)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE INDUSTRIAL PLANT EQUIP. CENTER ATTN: DIPEC-SSG	b. TELEPHONE *(Include Area Code)* (1) Commercial (901) 775-6989 (2) AUTOVON 683-6989
c. ADDRESS *(Include Zip Code)* 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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